

# L11: Inverse Percolation On Grids and $G(n,p)$ Networks

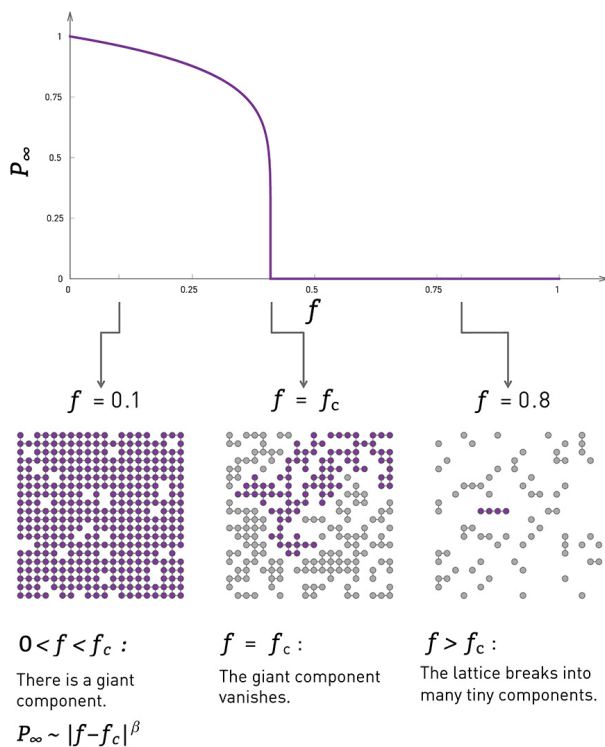


Image Source: **Network Science** by Albert-László Barabási .

Let's start with a process of network dynamics in which the topology changes over time due to node (or link) removals.

This happens often in practice as a result of failures (e.g., *Internet router malfunctions, removal of species from an ecosystem, mutations of genes in a regulatory network*) or deliberate attacks (e.g., *destroying specific generators in the power distribution network, arresting certain people in a terrorist network*).

Consider the simpler case in which instead of a network we have a two-dimensional grid, and a node is placed at every grid point. Two nodes are connected if they are in adjacent grid points.

In the context of random failures, we can select a fraction  $f$  of nodes and remove them from the grid, as shown in the visualization at the left. The y-axis in that visualization shows the probability that a randomly chosen node belongs in the largest connected component on the grid.

When  $f$  is low, the removals will not affect the grid much in the sense that most of the remaining nodes still form a single, large connected component (referred to as **giant component**).

As  $f$  increases however the originally connected grid will start “breaking apart” in small clusters of connected grid points.

Further, as the visualization shows, this transition is not gradual. Instead, there is a sudden drop in the size of the largest connected component as  $f$  approaches (from below) a critical threshold  $f_c$ .

At that point the **giant component** disappears, giving its place to a large number of smaller connected components.

In physics, this process is referred to as **inverse percolation** and it has been extensively studied for both regular grids as well as random  $G(n,p)$  networks.

The visualization at the right shows simulation results of a  $G(n,p)$  network with  $n=10,000$  nodes and average degree  $\bar{k} = 3$ .

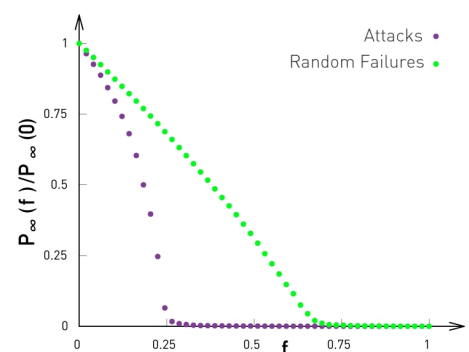
The y-axis shows, again, the probability that a node belongs in the largest connected component of the graph after we remove a fraction  $f$  of the nodes, normalized by the same probability when  $f=0$ .

In the case of random failures, note that the critical threshold  $f_c$  is about 0.65-0.70. After that point, the network is broken into small clusters of connected nodes and isolated nodes.

The same plot also shows what happens when we **attack** the highest-degree nodes of the network. We will come back to this curve a bit later in this Lesson.

On the next page, we will see an animation for power-law networks, in which the degree distribution has infinite variance. In that case, the process of random node removals behaves very different than the process of node attacks.

Then, after that animation, we will derive and show some mathematical results for the critical threshold –for the case of random failures as well as for the case of higher-degree node attacks.



## Food For Thought

Compare the inverse percolation process described here with what we studied in Lesson-3 about  $G(n,p)$  networks, namely the emergence of a giant connected component when the average degree exceeds one. In both cases we have a phase transition. What is common and what is different in the two processes?